

NUMERICAL IMPLEMENTATION AND ANALYSIS OF A CLASS OF METAL PLASTICITY MODELS COUPLED WITH DUCTILE DAMAGE

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SUMMARY

This paper deals with a class of rate-independent metal plasticity models which exhibit non-linear isotropic hardening, non-linear kinematic hardening (Chaboche–Marquis model) and ductile damage (Lemaitre–Chaboche model). The backward Euler scheme is used to integrate the rate constitutive relations. The non-linear equations obtained are solved by the Newton method. The consistent tangent operator is obtained by exact linearization of the algorithm. Despite the complexity of the constitutive equations, closed-form expressions are derived, without any approximations. Analytical, numerical and experimental results are presented and discussed.

KEY WORDS: damage; plasticity; finite elements

1. INTRODUCTION

In this paper, a class of rate-independent plasticity models exhibiting non-linear isotropic hardening, non-linear kinematic hardening (Chaboche–Marquis model) and ductile damage (Lemaitre–Chaboche model) is studied in detail, under isothermal and small strain conditions.

The non-linear kinematic hardening model—originally due to Armstrong and Frederick¹—has proven powerful in modelling cyclic plasticity for metals.

Continuum Damage Mechanisms (CDM), which represents a local approach to failure, is one of the most promising tools used to predict macro-crack initiation and propagation. In its ‘coupled’ approach, it can be described as follows.

The material behaviour is modelled by constitutive equations taking into account its progressive deterioration. The microscopic deterioration of a representative volume element (RVE) is modelled by a continuous (in space and time) variable defined at macro-scale and referred to as the *damage* variable. The ultimate phase of the damage evolution is detected by a local criterion and corresponds to the failure of the RVE, and hence to a macro-crack initiation. The spatial evolution of the completely damaged zone corresponds to the propagation of the macro-crack.

In Section 2, the class of constitutive models is presented and studied, with the numerical implementation in mind. Several results which are useful for numerical analysis are derived and discussed.

In Section 3, a strain-driven elastic predictor/plastic corrector return mapping algorithm² is developed. The rate constitutive relations are discretized in an incremental form corresponding to a fully implicit integration scheme.³ The non-linear system resulting from the backward Euler

discretization of the rate relations is solved iteratively by a Newton method. Without any approximation, it is proved that the solution can be updated explicitly.

In Section 4, the consistent tangent operator⁴ is computed by exact linearization of the algorithm. With no approximation, an explicit expression is found.

In Section 5, analytical solutions for loadings with constant triaxiality ratio are developed and discussed. The purpose of those developments is twofold. First, they contribute to a better understanding of the constitutive model, and second, they are used to check the implementation of the numerical procedure developed in Sections 3 and 4.

In Section 6, four numerical simulations are presented. The first two examples simulate ductile failure under uniaxial tension and simple shear, respectively. The numerical results are compared to analytical solutions. The third example predicts elasto-plastic behaviour under cyclic strain history. The last problem presents numerical and experimental results for a notched plate containing a macro-defect.

The non-damaged case is studied in Reference 5. However, as we shall see in this paper, adding a single (and scalar) damage variable adds a great deal of complexity to both the constitutive equations, and the numerical implementation. Nevertheless, as in Reference 5, this paper is self-contained; it enables efficient implementation of the constitutive equations presented in Section 2, using the algorithm developed in Sections 3 and 4.

It is now well known that the evolution of the damage variable induces strain softening which is the source of bifurcation and localization phenomena, which in turn are responsible for numerical troubles, such as pronounced mesh dependency of the results, in some cases. For information about those aspects, see Reference 6 and references therein.

Tensorial notation is used for convenience. Boldface symbols denote tensors, the order of which is indicated by the context. The symbol $\otimes$ denotes the tensorial product. The summation convention is used for Latin indices unless otherwise indicated. Dots and colons are used to indicate the following products:

$$(\mathbf{a} \cdot \mathbf{b})_{ij} = a_{ik} b_{kj}$$

$$\mathbf{a} : \mathbf{b} = a_{ij} b_{ji}$$

$$(\mathbf{c} : \mathbf{b})_{ij} = c_{ijkl} b_{lk}$$

$$(\mathbf{c} : \mathbf{d})_{ijkl} = c_{ijmn} d_{nmkl}$$

The inverse of a fourth order $\mathbf{c}$ is denoted by $\mathbf{c}^{-1}$ so that

$$\mathbf{c} : \mathbf{c}^{-1} = \mathbf{c}^{-1} : \mathbf{c} = \mathbf{I}$$

where $\mathbf{I}$ is the fourth-order identity tensor.

The following notations are also used:

$$J_2(\mathbf{a}) = [\frac{3}{2} \mathbf{a}^{\text{dev}} : \mathbf{a}^{\text{dev}}]^{1/2}$$

where $\mathbf{a}^{\text{dev}}$ is the deviatoric part of $\mathbf{a}$, i.e.

$$\mathbf{a}^{\text{dev}} = \mathbf{a} - \frac{1}{3} \text{tr}(\mathbf{a}) \mathbf{1} = \mathbf{I}^{\text{dev}} : \mathbf{a}$$

with $\mathbf{1}$ being the second-order identity tensor, $\text{tr}(\mathbf{a}) = a_{kk}$ and $\mathbf{I}^{\text{dev}} = \mathbf{I} - \frac{1}{3} \mathbf{1} \otimes \mathbf{1}$.

Throughout the paper, the following lemma will be very useful.

Lemma. If $\mathbf{c} = \mathbf{I} - (\mathbf{a} \otimes \mathbf{b})/h$ and $h_c = \mathbf{a}:\mathbf{b}$, then $\mathbf{c}$ is invertible if and only if $h \neq h_c$, and in this case:

$$\mathbf{c}^{-1} = \mathbf{I} + \frac{(\mathbf{a} \otimes \mathbf{b})}{h - h_c}$$

The proof for this lemma is straightforward.

2. CONSTITUTIVE EQUATIONS

We consider a family of metal plasticity models exhibiting non-linear isotropic hardening, non-linear kinematic hardening (Chaboche–Marquis model), and ductile damage (Lemaitre–Chaboche model). For additional information about these models, see References 7 and 8 and references therein.

The total strain is assumed to be the sum of an elastic part and a plastic part:

$$\boldsymbol{\varepsilon} = \boldsymbol{\varepsilon}^e + \boldsymbol{\varepsilon}^p \quad (1)$$

The stress, $\boldsymbol{\sigma}$, and the thermodynamic forces, $\mathbf{A}$, associated with the internal variables, $\mathbf{V}$, are derived from a specific free energy, ψ ($\boldsymbol{\varepsilon}^e$, $\mathbf{V}$), according to the state laws

$$\boldsymbol{\sigma} = \rho \frac{\partial \psi}{\partial \boldsymbol{\varepsilon}^e}, \quad \mathbf{A} = \rho \frac{\partial \psi}{\partial \mathbf{V}} \quad (2)$$

where ρ denotes the material density, and is assumed to be constant.

The internal variables $\mathbf{V}$ and their associated forces $\mathbf{A}$ are

$$\mathbf{V} = [r \quad \boldsymbol{\alpha} \quad D] \quad \text{and} \quad \mathbf{A} = [R \quad \mathbf{X} \quad (-Y)] \quad (3)$$

The scalar variable r models the isotropic hardening and the strain-like tensor $\boldsymbol{\alpha}$ models the kinematic hardening. The scalar variable R measures the radius of the yield surface in the space of deviatoric stresses and the stress-like tensor $\mathbf{X}$ measures the translation of the centre of that surface in the same space. The internal variable D models the damage, which is assumed to be isotropic; its associated variable is denoted by Y . The *extreme* values of the damage variable D are 0 for the sound material and 1 for a volume element with null stress carrying capability.

The specific free energy is defined by

$$\rho \psi = \frac{1}{2} \boldsymbol{\varepsilon}^e : (1 - D) \mathbf{E}_0 : \boldsymbol{\varepsilon}^e + \frac{1}{2} k_1 \boldsymbol{\alpha} : \boldsymbol{\alpha} + \Psi(r) \quad (4)$$

where $\mathbf{E}_0$ is the Hooke's operator of the sound material and k_1 is a material parameter.

The state laws (2) give

$$\boldsymbol{\sigma} = (1 - D) \mathbf{E}_0 : \boldsymbol{\varepsilon}^e \quad (5a)$$

$$R = \frac{d\Psi}{dr} \quad (5b)$$

$$\mathbf{X} = k_1 \boldsymbol{\alpha} \quad (5c)$$

$$Y = \frac{1}{2} \boldsymbol{\varepsilon}^e : \mathbf{E}_0 : \boldsymbol{\varepsilon}^e \quad (5d)$$

It appears that the variable Y associated with the damage D is the strain energy release rate.

An elastic domain within which no irreversibility is possible is defined by the yield function

$$f(\boldsymbol{\sigma}, \mathbf{A}; \mathbf{V}) \leq 0 \quad (6)$$

The evolution laws for the plastic strain and the internal variables are derived from a plastic potential, $F(\boldsymbol{\sigma}, \mathbf{A}; \mathbf{V})$, using the generalized normality rules

$$\dot{\boldsymbol{\varepsilon}}^p = \dot{\gamma} \frac{\partial F}{\partial \boldsymbol{\sigma}}, \quad \dot{\mathbf{V}} = -\dot{\gamma} \frac{\partial F}{\partial \mathbf{A}} \quad (7)$$

with the Kuhn–Tucker conditions

$$\dot{\gamma} \geq 0, \quad \dot{\gamma} f = 0 \quad \text{and} \quad \dot{\gamma} \dot{f} = 0 \quad (8)$$

The yield function and the plastic potential are chosen as

$$f = J_2(\tilde{\boldsymbol{\sigma}} - \mathbf{X}) - \sigma_y - R \leq 0 \quad (9)$$

$$F = f + \frac{k_2}{2k_1} \mathbf{X} : \mathbf{X} + F_D(Y; D) \quad (10)$$

where $\tilde{\boldsymbol{\sigma}} = \boldsymbol{\sigma}/(1 - D)$ is the so-called *effective stress*, σ_y is the initial yield stress and k_2 a material-dependent parameter.

Throughout the paper, it is assumed that Ψ , f and F are smooth functions, so that their partial derivatives which appear in the equations are well defined. We introduce the following notation:

$$\tilde{\mathbf{N}} = \frac{\partial f}{\partial \tilde{\boldsymbol{\sigma}}} = \frac{3}{2} \frac{(\tilde{\mathbf{s}} - \mathbf{X})}{J_2(\tilde{\boldsymbol{\sigma}} - \mathbf{X})} = -\frac{\partial f}{\partial \mathbf{X}} \quad (11)$$

where $\tilde{\mathbf{s}}$ is the deviatoric part of $\tilde{\boldsymbol{\sigma}}$. If there is no damage, the tensor $\tilde{\mathbf{N}}$ is the normal to the yield surface in the stress space. The evolution laws (7) give

$$\dot{\boldsymbol{\varepsilon}}^p = \dot{\gamma} \frac{\tilde{\mathbf{N}}}{1 - D} \quad (12a)$$

$$\dot{r} = \dot{\gamma} \quad (12b)$$

$$\dot{\boldsymbol{\alpha}} = \dot{\gamma} \left(\tilde{\mathbf{N}} - \frac{k_2}{k_1} \mathbf{X} \right) \quad (12c)$$

$$\dot{D} = \dot{\gamma} \frac{\partial F_D}{\partial Y} \quad (12d)$$

Note that the tensors $\boldsymbol{\varepsilon}^p$, $\tilde{\mathbf{N}}$ and $\mathbf{X}$ are *deviatoric*. Since $\tilde{\mathbf{N}} : \tilde{\mathbf{N}} = \frac{3}{2}$, it is easily shown that the accumulated plastic strain rate $\dot{p}$ defined by

$$\dot{p} = \sqrt{\frac{2}{3} \dot{\boldsymbol{\varepsilon}}^p : \dot{\boldsymbol{\varepsilon}}^p} \quad (13)$$

is related to the plastic multiplier $\dot{\gamma}$ by

$$\dot{\gamma} = \dot{r} = (1 - D) \dot{p} \quad (14)$$

When there is damage, the internal variable r which models the isotropic hardening is *not* the accumulated plastic strain p .

To summarize, the model can be described by the following set of equations:

$$\boldsymbol{\sigma} = (1 - D)\mathbf{E}_0 : \boldsymbol{\varepsilon}^e \quad (15a)$$

$$f = J_2(\boldsymbol{\sigma} - \mathbf{X}) - \sigma_y - R(r) \leq 0 \quad (15b)$$

$$\dot{\boldsymbol{\varepsilon}}^p = \dot{p}\tilde{\mathbf{N}} \quad (15c)$$

$$\dot{\mathbf{X}} = (1 - D)(k_1\dot{\boldsymbol{\varepsilon}}^p - k_2\mathbf{X}\dot{p}) \quad (15d)$$

$$\dot{D} = \dot{r} \frac{\partial F_D}{\partial Y} \quad (15e)$$

$$\dot{r} = (1 - D)\dot{p} \geq 0, \quad \dot{r}f = 0 \quad \text{and} \quad \dot{r}\dot{f} = 0 \quad (15f)$$

In practice, the isotropic hardening potential is chosen as

$$\Psi(r) = \frac{k}{n+1} r^{n+1} \quad \text{or} \quad \Psi(r) = R_\infty \left[r + \frac{1}{n} \exp(-nr) - \frac{1}{n} \right] \quad (16)$$

which gives a power law or an exponential law (with saturation):

$$R = kr^n \quad \text{or} \quad R = R_\infty [1 - \exp(-nr)] \quad (17)$$

where k , n and R_∞ are material parameters.

We now examine the kinematic hardening evolution (15d). If $k_2 = 0$ and there is no damage, we retrieve a linear kinematic hardening model. If $k_2 \neq 0$ a non-linear term (a recall term) is introduced which allows a good modelling of cyclic plasticity.⁹⁻¹⁴ Note also that for a uniaxial tension under monotonically increasing loading, equation (15d) can be integrated in time to give (when $k_2 \neq 0$)

$$X = \frac{k_1}{k_2} [1 - \exp(-k_2 r)] \quad (18)$$

This shows that the non-linear kinematic hardening saturates for large values of plastic strain ($X \rightarrow k_1/k_2$ when $r \rightarrow \infty$).

In practice, the damage potential $F_D(Y; D)$ is chosen as

$$F_D(Y; D) = \frac{S_0}{2} \left(\frac{Y}{S_0} \right)^2 \frac{1}{1 - D} \quad (19)$$

which gives the damage evolution as

$$\dot{D} = \frac{Y}{S_0} \dot{p} \geq 0 \quad (20)$$

where S_0 is a material parameter. Note that it is usually observed that damage does not develop under a certain plastic threshold: $\dot{D} = 0$ if $p \leq p_D$, where p_D can be related to the energy stored in the material.¹⁵

In the remainder of this paper, we will assume that the linear elasticity is isotropic. We then have $\mathbf{E}_0 = 2G_0\mathbf{I}^{\text{dev}} + K_0\mathbf{1} \otimes \mathbf{1}$, where G_0 and K_0 are the elastic shear and bulk moduli for the sound material, respectively,

$$G_0 = \frac{E_0}{2(1 + \nu)} \quad \text{and} \quad K_0 = \frac{E_0}{3(1 - 2\nu)} \quad (21)$$

Recalling the definition of Y , equation (5d), it can be shown that

$$Y = \frac{[J_2(\tilde{\sigma})]^2}{2E_0} R_v, \quad \text{with } R_v = \frac{2}{3}(1 + \nu) + 3(1 - 2\nu) \left[\frac{\sigma_H}{J_2(\tilde{\sigma})} \right]^2 \quad (22)$$

where σ_H is the hydrostatic stress: $\sigma_H = \sigma_{kk}/3$.

The variable Y and consequently the damage rate $\dot{D}$ depend strongly on the triaxiality ratio $\sigma_H/J_2(\tilde{\sigma})$ which is known to be a main feature in failure. It has been suggested¹⁶ that for ductile damage, the yield function should depend on the hydrostatic stress. This is not the case for the damage model studied here, where the yield function, equation (9), is always deviatoric. However, because of the particular form of the damage model, equations (20) and (22), and because of the way elasto-plasticity and damage equations are coupled together, equations (15), this damage model simulates reasonably well the ductile behaviour of several metals (see References 7, 8, 15 and references therein). Also, several ductile fracture models, including Gurson's damage model¹⁷ — which uses a pressure-sensitive yield function — and Lemaitre–Chaboche model have been compared in Reference 18. The latter model was found to give good predictions of ductile behaviour. In particular, for a cavity growth problem, it gave a result similar to the classical solution of Rice and Tracey.¹⁹

The inequality of Clausius–Duhem requires the dissipation, $\mathcal{D}$, to be positive:²⁰

$$\mathcal{D} = \boldsymbol{\sigma} : \dot{\boldsymbol{\epsilon}}^p - \mathbf{A} \cdot \dot{\mathbf{V}} \geq 0 \quad (23)$$

For the model under consideration, this inequality becomes

$$\mathcal{D} = \boldsymbol{\sigma} : \dot{\boldsymbol{\epsilon}}^p - R\dot{r} - \mathbf{X} : \dot{\boldsymbol{\alpha}} + Y\dot{D} \geq 0 \quad (24)$$

After some algebraic manipulation, this requirement takes the simple form

$$\mathcal{D} = \left(\sigma_y + \frac{k_2}{k_1} \mathbf{X} : \mathbf{X} + Y \frac{\partial F_D}{\partial Y} \right) \dot{r} \geq 0 \quad (25)$$

and is always satisfied.

2.1. Plastic multiplier $\dot{r}$

When $\dot{r} > 0$, the consistency condition $\dot{f} = 0$ holds. After some algebraic manipulation, it is found that

$$\dot{f} = 2G_0 \tilde{\mathbf{N}} : \dot{\boldsymbol{\epsilon}} - h\dot{p} \quad (26)$$

where

$$h = 3G_0 + (1 - D) \left(\frac{dR}{dr} + \frac{3}{2}k_1 - k_2 \tilde{\mathbf{N}} : \mathbf{X} \right) \quad (27)$$

which gives

$$\dot{p} = \frac{\dot{r}}{1 - D} = \frac{2G_0 \tilde{\mathbf{N}} : \dot{\boldsymbol{\epsilon}}}{h} \quad (28)$$

Note that the modulus h has to be *positive*. Mechanical interpretations of this condition are given in References 21 and 22. If $k_2 = 0$ (kinematic hardening is linear, or absent) the condition $h > 0$ is always satisfied.

2.2. Tangent modulus L

The tangent modulus, L , is defined by the rate relation: $\dot{\boldsymbol{\sigma}} = \mathbf{L} : \dot{\boldsymbol{\varepsilon}}$. It is found that $\mathbf{L}$ has two possible expressions, $\mathbf{E}$ and $\mathbf{H}$.

- (i) If $f < 0$ (elasticity) or $f = 0$ and $\dot{f} < 0$ (elastic unloading), then $\dot{\gamma} = 0$ and $\mathbf{L} = \mathbf{E}$.
- (ii) If $f = 0$ and $\dot{f} = 0$ (plastic loading), then $\dot{\gamma} > 0$ and $\mathbf{L} = \mathbf{H}$.

Using the results of Section 2.1, it is found that elastic unloading corresponds to the quantity $(2G_0\tilde{\mathbf{N}} : \dot{\boldsymbol{\varepsilon}})/h$ being negative, while plastic loading corresponds to that same quantity being positive.

The elastic expression of the tangent operator is

$$\mathbf{E} = (1 - D)\mathbf{E}_0 \quad (29)$$

For classical models, $\mathbf{E}$ is Hooke's operator, but for the damage model considered in this paper, $\mathbf{E}$ is the elasticity tensor of the *damaged* material. Equation (29) also shows that the damage variable can be measured by the decay of Young's modulus ($D = 1 - E/E_0$).

The plastic expression of the tangent operator is found to be

$$\mathbf{H} = (1 - D) \left[\mathbf{E}_0 - \frac{1}{h} \left(2G_0\tilde{\mathbf{N}} + \frac{\partial F_D}{\partial Y} \tilde{\boldsymbol{\sigma}} \right) \otimes (2G_0\tilde{\mathbf{N}}) \right] \quad (30)$$

Due to the coupling with damage, $\mathbf{H}$ is *not* symmetric. Note that $\mathbf{H}$ can be written in the following form:

$$\mathbf{H} = \mathbf{E} - \frac{(\mathbf{E} : \mathbf{a}) \otimes (\mathbf{b} : \mathbf{E})}{h} \quad (31)$$

where

$$\mathbf{a} \equiv \tilde{\mathbf{N}} + \frac{\partial F_D}{\partial Y} \boldsymbol{\varepsilon}^e \quad \text{and} \quad \mathbf{b} \equiv \frac{\tilde{\mathbf{N}}}{1 - D} \quad (32)$$

2.3. Hardening modulus h^*

From the above relations,

$$\mathbf{b} : \dot{\boldsymbol{\sigma}} = \dot{p}(h - \mathbf{b} : \mathbf{E} : \mathbf{a}) \quad (33)$$

The *hardening modulus* h^* is defined by

$$h^* \equiv (h - \mathbf{b} : \mathbf{E} : \mathbf{a}) = \frac{\mathbf{b} : \dot{\boldsymbol{\sigma}}}{\dot{p}} \quad (34)$$

Using the lemma given in Section 1, the tangent modulus, $\mathbf{H}$, is found to be *invertible* if and only if the hardening modulus, h^* , is non zero, and in this case,

$$\mathbf{H}^{-1} = \mathbf{E}^{-1} + \frac{\mathbf{a} \otimes \mathbf{b}}{h^*} \quad (35)$$

These results are *useful* for numerical analysis and *simple* to implement.

Relation (34b) gives for this model the hardening modulus as

$$h^* = \frac{\mathbf{b} : \dot{\boldsymbol{\sigma}}}{\dot{p}} = \frac{\dot{\boldsymbol{\sigma}} : \dot{\boldsymbol{\varepsilon}}^p}{\dot{p}} \quad (36)$$

Since $\mathbf{b} = \partial f / \partial \boldsymbol{\sigma}$, when h^* is positive or negative, we have *hardening* or *softening*, respectively. Definition (34a) together with the expressions of h , $\mathbf{a}$ and $\mathbf{b}$ leads to

$$h^* = (1 - D) \left(\frac{dR}{dr} + \frac{3}{2} k_1 - k_2 \tilde{\mathbf{N}} : \mathbf{X} \right) - \frac{\partial F_D}{\partial Y} \tilde{\mathbf{N}} : \tilde{\boldsymbol{\sigma}} \quad (37)$$

3. FULLY IMPLICIT INTEGRATION WITH EXPLICIT UPDATING

The method used is a strain-driven algorithm: at each ‘time’ (or time-like parameter) increment, and for a given total strain increment, one knows the values of the stresses and the other material model variables at the beginning of the increment (t_n) and they have to be updated at the end of the increment (t_{n+1}). Plane stress is a special case for which the out-of-plane strain is unknown: a modification of the following algorithm to deal with this issue is presented in the appendix.

The subscript $(n + 1)$ will be omitted and all the variables which do not contain the subscript n are computed at t_{n+1} . Also, a notation is used to designate an increment: $\Delta(\cdot) = (\cdot)_{n+1} - (\cdot)_n$. We first assume that the increment is entirely elastic, i.e. $\gamma = \gamma_n$. We then have $\mathbf{V} = \mathbf{V}_n$ and $\boldsymbol{\sigma}$ and $\mathbf{A}$ are found using the state laws (2). If this so-called elastic predictor satisfies the yield condition $f \leq 0$, then the assumption is valid and the computation for this time increment ends. If $f > 0$, this elastic state is ‘corrected’ as explained in the following in order to find the ‘plastic’ solution.

For the constitutive model under study, the elastic predictor phase is defined by the following trial stresses:

$$\tilde{\boldsymbol{\sigma}}^{\text{trial}} = \mathbf{E}_0 : (\boldsymbol{\varepsilon} - \boldsymbol{\varepsilon}_n^p), \quad \boldsymbol{\sigma}^{\text{trial}} = (1 - D_n) \tilde{\boldsymbol{\sigma}}^{\text{trial}} \quad (38)$$

All other ‘plastic’ variables are equal to their values at t_n . If this trial state does not satisfy the condition $f \leq 0$, then the solution at t_{n+1} is found as follows.

First, we note the following relation between $\tilde{\boldsymbol{\sigma}}$ and $\tilde{\boldsymbol{\sigma}}^{\text{trial}}$.

$$\tilde{\boldsymbol{\sigma}} = \tilde{\boldsymbol{\sigma}}^{\text{trial}} - 2G_0 \Delta \boldsymbol{\varepsilon}^p \quad (39)$$

If the true stresses, $\boldsymbol{\sigma}$, are used instead of the effective stresses, $\tilde{\boldsymbol{\sigma}}$ such a simple relation *cannot* be found.

Second, noting that

$$\text{tr } \tilde{\boldsymbol{\sigma}} = \text{tr } \tilde{\boldsymbol{\sigma}}^{\text{trial}} = 3K_0 \text{tr } \boldsymbol{\varepsilon} \quad (40)$$

only the deviatoric part of (39) need to be considered. Time discretization of constitutive equations (15) using the backward Euler scheme gives the following set of equations:

$$\tilde{\mathbf{s}} = \tilde{\mathbf{s}}^{\text{trial}} - 2G_0 \Delta \boldsymbol{\varepsilon}^p \quad (41a)$$

$$f \equiv J_2(\tilde{\boldsymbol{\sigma}} - \mathbf{X}) - \sigma_y - R(r) = 0 \quad (41b)$$

$$\Delta \boldsymbol{\varepsilon}^p = \tilde{\mathbf{N}} \Delta p \quad (41c)$$

$$\Delta \mathbf{X} = (1 - D)(k_1 \Delta \boldsymbol{\varepsilon}^p - k_2 \mathbf{X} \Delta p) \quad (41d)$$

$$\Delta D = y(\tilde{\mathbf{s}}) \Delta p \quad (41e)$$

$$\Delta r = (1 - D) \Delta p \quad (41f)$$

where the notation $y \equiv Y/S_0$ was introduced for convenience. Noting that $\sigma_H/J_2(\boldsymbol{\sigma}) = \tilde{\sigma}_H/J_2(\tilde{\boldsymbol{\sigma}})$, and using (22) and (40), it appears that y only depends on $\tilde{\mathbf{s}}$ (through $J_2(\tilde{\boldsymbol{\sigma}})$). After substitution of $\Delta \boldsymbol{\varepsilon}^p$ and Δp by their expressions, it is clear that the problem is reduced to finding the four

unknowns $\tilde{s}$, r , $\mathbf{X}$, D which satisfy the following system of four equations:

$$\mathbf{k}_{\tilde{s}} \equiv \tilde{s} - \tilde{s}^{\text{trial}} + 2G_0 \tilde{\mathbf{N}} \frac{\Delta r}{1 - D} = \mathbf{0} \quad (42a)$$

$$f \equiv J_2(\tilde{\boldsymbol{\sigma}} - \mathbf{X}) - \sigma_y - R(r) = 0 \quad (42b)$$

$$\mathbf{k}_{\mathbf{X}} \equiv \Delta \mathbf{X} - (k_1 \tilde{\mathbf{N}} - k_2 \mathbf{X}_n) \frac{\Delta r}{1 + k_2 \Delta r} = \mathbf{0} \quad (42c)$$

$$k_D \equiv \Delta D - y(\tilde{s}) \frac{\Delta r}{1 - D} = 0 \quad (42d)$$

Because of the damage coupling, it is not possible to reduce the problem to finding two unknowns, r and $(\tilde{s} - \mathbf{X})$, as in Reference 5. Newton's method is used to solve the non-linear system of equations (42). For each iteration (s) of the Newton scheme, we have

$$\mathbf{k}_{\tilde{s}} + 2G_0 \tilde{\mathbf{N}} c_p + \mathbf{c}_{\tilde{s}} + 2G_0 \Delta p \frac{\partial \tilde{\mathbf{N}}}{\partial \tilde{s}} : (\mathbf{c}_{\tilde{s}} - \mathbf{c}_{\mathbf{X}}) = \mathbf{0} \quad (43a)$$

$$f + \tilde{\mathbf{N}} : (\mathbf{c}_{\tilde{s}} - \mathbf{c}_{\mathbf{X}}) - \frac{dR}{dr} c_r = 0 \quad (43b)$$

$$\mathbf{k}_{\mathbf{X}} + \mathbf{c}_{\mathbf{X}} - (k_1 \tilde{\mathbf{N}} - k_2 \mathbf{X}_n) \frac{c_r}{(1 + k_2 \Delta r)^2} - \frac{k_1 \Delta r}{1 + k_2 \Delta r} \frac{\partial \tilde{\mathbf{N}}}{\partial \tilde{s}} : (\mathbf{c}_{\tilde{s}} - \mathbf{c}_{\mathbf{X}}) = \mathbf{0} \quad (43c)$$

$$k_D + c_D - y c_p - \Delta p \frac{\partial y}{\partial \tilde{s}} : \mathbf{c}_{\tilde{s}} = 0 \quad (43d)$$

where $\Delta p = \Delta r / (1 - D)$ and $c_p \equiv (c_r + \Delta p c_D) / (1 - D)$. The notation $c_{(\cdot)} \equiv (\cdot)_{s+1} - (\cdot)_s$ designates an iterative 'correction'. Each variable that does not have the subscript n is computed at 'time' t_{n+1} and the plastic iteration (s).

The general procedure involves matrix inversions or—in less general cases^{23–25}—the resolution of linear systems for each local iteration (s). Although those computations are carried out at the integration point level, their cost may become prohibitive. Despite the complexity of the constitutive equations considered in this paper, we shall show that the iterative updating is *explicit*. Unlike the modified 'cutting plane algorithm'^{2,26} *no approximations* will be made.

Combining equations (43a) and (43c) gives

$$\mathbf{B} : (\mathbf{c}_{\tilde{s}} - \mathbf{c}_{\mathbf{X}}) = - \left[\mathbf{k}_{\tilde{s}} - \mathbf{k}_{\mathbf{X}} + 2G_0 \tilde{\mathbf{N}} c_p + (k_1 \tilde{\mathbf{N}} - k_2 \mathbf{X}_n) \frac{c_r}{(1 + k_2 \Delta r)^2} \right] \quad (44)$$

where

$$\mathbf{B} \equiv \mathbf{I} + \left(2G_0 \Delta p + \frac{k_1 \Delta r}{1 + k_2 \Delta r} \right) \frac{\partial \tilde{\mathbf{N}}}{\partial \tilde{s}} = (1 + \frac{3}{2}g) \mathbf{I} - g \tilde{\mathbf{N}} \otimes \tilde{\mathbf{N}} \quad (45)$$

$$\frac{\partial \tilde{\mathbf{N}}}{\partial \tilde{s}} = \frac{\frac{3}{2} \mathbf{I} - \tilde{\mathbf{N}} \otimes \tilde{\mathbf{N}}}{J_2(\tilde{\boldsymbol{\sigma}} - \mathbf{X})} = - \frac{\partial \tilde{\mathbf{N}}}{\partial \mathbf{X}} \quad (46)$$

$$g \equiv \frac{2G_0 \Delta p + k_1 \Delta r / (1 + k_2 \Delta r)}{J_2(\tilde{\boldsymbol{\sigma}} - \mathbf{X})} > 0 \quad (47)$$

Using the lemma given in Section 1, it is found that $\mathbf{B}$ is invertible if and only if $1 + (\frac{3}{2})g \neq 0$, which is always verified for an acceptable iterative approximation. The inverse of $\mathbf{B}$ is given by

$$\mathbf{B}^{-1} = \frac{\mathbf{I} + g\tilde{\mathbf{N}} \otimes \tilde{\mathbf{N}}}{1 + (\frac{3}{2})g} \quad (48)$$

Using these results, equation (44) is equivalent to the following equation:

$$\begin{aligned} \mathbf{c}_s - \mathbf{c}_x = & \frac{-1}{1 + (\frac{3}{2})g} [\mathbf{k}_s - \mathbf{k}_x + g\tilde{\mathbf{N}} : (\mathbf{k}_s - \mathbf{k}_x) \tilde{\mathbf{N}}] - \left[2G_0 c_p + \frac{k_1}{(1 + k_2 \Delta r)^2} c_r \right] \tilde{\mathbf{N}} \\ & + \frac{1}{(1 + \frac{3}{2}g)(1 + k_2 \Delta r)^2} [\mathbf{X}_n + g(\tilde{\mathbf{N}} : \mathbf{X}_n) \tilde{\mathbf{N}}] c_r \end{aligned} \quad (49)$$

Noting that $\partial \tilde{\mathbf{N}} / \partial \tilde{\mathbf{s}} : \tilde{\mathbf{N}} = \mathbf{0}$, the following result follows easily:

$$\begin{aligned} \frac{\partial \tilde{\mathbf{N}}}{\partial \tilde{\mathbf{s}}} : (\mathbf{c}_s - \mathbf{c}_x) = & \frac{-3}{2} \frac{1}{J_2(\tilde{\mathbf{s}} - \tilde{\mathbf{X}})} \frac{1}{(1 + \frac{3}{2}g)} \left\{ \mathbf{k}_s - \mathbf{k}_x - \frac{2}{3} \tilde{\mathbf{N}} : (\mathbf{k}_s - \mathbf{k}_x) \tilde{\mathbf{N}} \right. \\ & \left. - \frac{k_2}{(1 + k_2 \Delta r)^2} \left[\mathbf{X}_n - \frac{2}{3} (\tilde{\mathbf{N}} : \mathbf{X}_n) \tilde{\mathbf{N}} \right] c_r \right\} \end{aligned} \quad (50)$$

From (43a) an expression of $\mathbf{c}_s$ as a function of c_r and c_p is found:

$$\mathbf{c}_s = -\mathbf{k}_s - 2G_0 \tilde{\mathbf{N}} c_p + \frac{3G_0}{J_2(\tilde{\mathbf{s}} - \mathbf{X})} \frac{\Delta p}{(1 + \frac{3}{2}g)} \{ \dots \} \quad (51)$$

Using (43d), an expression of c_D as a function of c_r and c_p is found:

$$c_D = -k_D + y c_p + \Delta p \frac{\partial y}{\partial \tilde{\mathbf{s}}} : \left[-\mathbf{k}_s - 2G_0 \tilde{\mathbf{N}} c_p + \frac{3G_0}{J_2(\tilde{\mathbf{s}} - \mathbf{X})} \frac{\Delta p}{(1 + \frac{3}{2}g)} \{ \dots \} \right] \quad (52)$$

In both (51) and (52), the expression between brackets is given by (50).

Using (43b), an expression of c_p as a function of c_r is found:

$$c_p = \frac{1}{3G_0} \left\{ f - \tilde{\mathbf{N}} : (\mathbf{k}_s - \mathbf{k}_x) - \left[\frac{dR}{dr} + \frac{\frac{3}{2}k_1 - k_2(\tilde{\mathbf{N}} : \mathbf{X}_n)}{(1 + k_2 \Delta r)^2} \right] c_r \right\} \quad (53)$$

Finally, the definition of c_p is used to find the expression of c_r :

$$c_r = \frac{\mathcal{N}}{h^{\text{alg}}} \quad (54)$$

The numerator is given by the following expression:

$$\begin{aligned} \mathcal{N} = & \left[(1 - D) - y \Delta p + 2G_0 (\Delta p)^2 \frac{\partial y}{\partial \tilde{\mathbf{s}}} : \tilde{\mathbf{N}} \right] [f - \tilde{\mathbf{N}} : (\mathbf{k}_s - \mathbf{k}_x)] \\ & + 3G_0 \left[\Delta p k_D + (\Delta p)^2 \frac{\partial y}{\partial \tilde{\mathbf{s}}} : \mathbf{k}_s \right] \\ & - 3G_0 (\Delta p)^2 \frac{3G_0}{J_2(\tilde{\mathbf{s}} - \mathbf{X})} \frac{\Delta p}{(1 + \frac{3}{2}g)} \frac{\partial y}{\partial \tilde{\mathbf{s}}} : [\mathbf{k}_s - \mathbf{k}_x - \frac{2}{3} \tilde{\mathbf{N}} : (\mathbf{k}_s - \mathbf{k}_x) \tilde{\mathbf{N}}] \end{aligned} \quad (55)$$

The denominator is given by the following expression:

$$h^{\text{alg}} = 3G_0 + \left[(1 - D) - y\Delta p + 2G_0(\Delta p)^2 \frac{\partial y}{\partial \tilde{\mathbf{s}}} : \tilde{\mathbf{N}} \right] \left[\frac{dR}{dr} + \frac{\frac{3}{2}k_1 - k_2(\tilde{\mathbf{N}} : \mathbf{X}_n)}{(1 + k_2\Delta r)^2} \right] \\ - 3G_0(\Delta p)^2 \frac{3G_0}{J_2(\tilde{\boldsymbol{\sigma}} - \mathbf{X})} \frac{\Delta p}{(1 + \frac{3}{2}g)} \frac{k_2}{(1 + k_2\Delta r)^2} \frac{\partial y}{\partial \tilde{\mathbf{s}}} : [\mathbf{X}_n - \frac{2}{3}(\tilde{\mathbf{N}} : \mathbf{X}_n) \tilde{\mathbf{N}}] \quad (56)$$

Note that if the plasticity increment is *very small*, i.e. if $\Delta p \rightarrow 0$, then $h^{\text{alg}} \rightarrow h$, where h is the ‘continuum’ modulus given by equation (27).

3.1. Corrections over the elastic predictor

For the elastic predictor, the increments of r , $\mathbf{X}$, D and p are zero. All the residual functions (42), except f , also vanish. The corrections over the elastic predictor have the following simple expressions:

$$c_r = \frac{(1 - D_n)f^{\text{trial}}}{h^{\text{trial}}} \quad (57)$$

$$c_p = \frac{c_r}{1 - D_n} \quad (58a)$$

$$\mathbf{c}_{\tilde{\mathbf{s}}} = -2G_0 \tilde{\mathbf{N}}^{\text{trial}} c_p \quad (58b)$$

$$\mathbf{c}_{\mathbf{X}} = (k_1 \tilde{\mathbf{N}}^{\text{trial}} - k_2 \mathbf{X}_n) c_r \quad (58c)$$

$$c_D = y^{\text{trial}} c_p \quad (58d)$$

Results (58) show that the corrections over the elastic predictor correspond to a *discretization* of the rate constitutive equations (i.e. by replacing $\dot{p}$ with c_p/dt , $\dot{r}$ with c_r/dt , $\dot{D}$ with c_D/dt , $\dot{\tilde{\mathbf{s}}}$ with $\mathbf{c}_{\tilde{\mathbf{s}}}/dt$ and $\dot{\mathbf{X}}$ with $\mathbf{c}_{\mathbf{X}}/dt$).

3.2. Summary of the algorithm

The time-integration algorithm can be summarized as follows.

Elastic predictor. Compute $\tilde{\boldsymbol{\sigma}}^{\text{trial}}$ and $\boldsymbol{\sigma}^{\text{trial}}$ from equations (38).

If $f^{\text{trial}} \leq 0$, then update the solution: $\boldsymbol{\sigma} = \boldsymbol{\sigma}^{\text{trial}}$, $r = r_n$, $\mathbf{X} = \mathbf{X}_n$, $p = p_n$, $D = D_n$, and exit.

Else:

 Compute corrections $c_{(\cdot)}$ over the elastic predictor, as in Section 3.1.

 Loop over plastic iterations:

 Update variables $(\cdot)$ according to: $(\cdot) \leftarrow (\cdot) + c_{(\cdot)}$.

 Compute residuals from equations (42).

 If tolerance criteria on residuals are satisfied, then exit the loop.

 Else: compute corrections $c_{(\cdot)}$ as follows

 Find c_r from equation (54)

 Find c_p from equation (53)

 Find $\mathbf{c}_{\tilde{\mathbf{s}}}$ from equation (51)

 Find $\mathbf{c}_{\tilde{\mathbf{s}}} - \mathbf{c}_{\mathbf{X}}$ from equation (49)

 Update $\mathbf{c}_{\mathbf{X}} = \mathbf{c}_{\tilde{\mathbf{s}}} - (\mathbf{c}_{\tilde{\mathbf{s}}} - \mathbf{c}_{\mathbf{X}})$

 Find c_D from equation (52).

 End if

End loop

End if

3.3. Study of the non-damaged case

If damage is inactive during the time increment under consideration, then the following simplifications hold: $\Delta D = 0$, $D = D_n$, $y = 0$, $k_D = 0$, $\partial y / \partial \tilde{s} = 0$ and $c_D = 0$. If, in addition, D_n is zero, then $p = r$, $\tilde{s} = s$. In that case, it is easily verified that one needs to compute corrections for two variables only: r and $(s - X)$. The correction over r is given by

$$c_r = \frac{f - \mathbf{N} : (\mathbf{k}_{\tilde{s}} - \mathbf{k}_X)}{3G_0 + \frac{dR}{dr} + \frac{\frac{3}{2}k_1 - k_2(\mathbf{N} : \mathbf{X}_n)}{(1 + k_2\Delta r)^2}} \quad (59)$$

The correction over $(s - X)$ is given by (49). Equations (59) and (49) are identical to equations (49) and (55) in Reference 5.

4. CONSISTENT TANGENT OPERATOR

When the local integration algorithm described in Section 3 has converged, the corresponding consistent tangent operator $\mathbf{L}^{\text{alg}}$ is determined by letting all the variables, including the total strain tensor $\boldsymbol{\varepsilon}$, vary slightly around the converged solution at t_{n+1} , so that $\delta \boldsymbol{\sigma} = \mathbf{L}^{\text{alg}} : \delta \boldsymbol{\varepsilon}$. If the increment is entirely elastic (i.e. if the elastic predictor is the solution), then $\mathbf{L}^{\text{alg}} = (1 - D_n)\mathbf{E}_0$; otherwise $\mathbf{L}^{\text{alg}} = \mathbf{H}^{\text{alg}}$, where the consistent tangent operator $\mathbf{H}^{\text{alg}}$ is found as described in this section.

Differentiation of the incremental relations (41) gives

$$\delta \tilde{s} = 2G_0 \mathbf{I}^{\text{dev}} : \delta \boldsymbol{\varepsilon} - 2G_0 \tilde{\mathbf{N}} \delta p - 2G_0 \Delta p \frac{\partial \tilde{\mathbf{N}}}{\partial \tilde{s}} : (\delta \tilde{s} - \delta \mathbf{X}) \quad (60a)$$

$$\tilde{\mathbf{N}} : (\delta \tilde{s} - \delta \mathbf{X}) - \frac{dR}{dr} \delta r = 0 \quad (60b)$$

$$\delta \mathbf{X} = (k_1 \tilde{\mathbf{N}} - k_2 \mathbf{X}_n) \frac{\delta r}{(1 + k_2 \Delta r)^2} + \frac{k_1 \Delta r}{1 + k_2 \Delta r} \frac{\partial \tilde{\mathbf{N}}}{\partial \tilde{s}} : (\delta \tilde{s} - \delta \mathbf{X}) \quad (60c)$$

$$\delta D = y \delta p + \Delta p \frac{\partial y}{\partial \tilde{\boldsymbol{\sigma}}} : \delta \tilde{\boldsymbol{\sigma}} \quad (60d)$$

$$\delta r = (1 - D) \delta p - \Delta p \delta D \quad (60e)$$

All the results of Section 3 can be used if we note that

$$\frac{\partial y}{\partial \tilde{\boldsymbol{\sigma}}} : \delta \tilde{\boldsymbol{\sigma}} = \frac{\partial y}{\partial \tilde{s}} : \delta \tilde{s} + \frac{\partial y}{\partial \text{tr } \tilde{\boldsymbol{\sigma}}} \delta(\text{tr } \tilde{\boldsymbol{\sigma}}) \quad (61)$$

and

$$\delta(\text{tr } \tilde{\boldsymbol{\sigma}}) = 3K_0 \mathbf{1} : \delta \boldsymbol{\varepsilon} \quad (62)$$

and if we consider the following *analogy* relations:

$$\mathbf{k}_{\tilde{s}} \leftrightarrow -2G_0 \mathbf{I}^{\text{dev}} : \delta \boldsymbol{\varepsilon} \quad (63a)$$

$$f \leftrightarrow 0 \quad (63b)$$

$$\mathbf{k}_X \leftrightarrow 0 \quad (63c)$$

$$k_D \leftrightarrow -\Delta p \frac{\partial y}{\partial \text{tr } \tilde{\boldsymbol{\sigma}}} 3K_0 \mathbf{1} : \delta \boldsymbol{\varepsilon} \quad (63d)$$

By analogy with (54), δr is found to be

$$\delta r = \frac{\mathbf{n}^{\text{alg}} : \delta \boldsymbol{\varepsilon}}{h^{\text{alg}}} \quad (64)$$

where

$$\begin{aligned} \mathbf{n}^{\text{alg}} = & \left[(1 - D) - y\Delta p + 2G_0(\Delta p)^2 \frac{\partial y}{\partial \tilde{\mathbf{s}}} : \tilde{\mathbf{N}} \right] (2G_0 \tilde{\mathbf{N}}) \\ & - 3G_0(\Delta p)^2 \left[\mathbf{E}_0 - (2G_0)^2 \frac{\Delta p}{(1 + \frac{3}{2}g)} \frac{\partial^2 f}{\partial \tilde{\boldsymbol{\sigma}} \partial \tilde{\boldsymbol{\sigma}}} \right] : \frac{\partial y}{\partial \tilde{\boldsymbol{\sigma}}} \end{aligned} \quad (65)$$

with

$$\frac{\partial^2 f}{\partial \tilde{\boldsymbol{\sigma}} \partial \tilde{\boldsymbol{\sigma}}} = \frac{\frac{3}{2}\mathbf{I}^{\text{dev}} - \tilde{\mathbf{N}} \otimes \tilde{\mathbf{N}}}{J_2(\tilde{\boldsymbol{\sigma}} - \mathbf{X})} \quad (66)$$

Note that if the plasticity increment is *very small*, i.e. if $\Delta p \rightarrow 0$, then $\delta r \rightarrow \dot{r} dt$, where $\dot{r}$ is the ‘continuum’ plastic multiplier given by equation (28).

The variation of p is found by analogy with (53):

$$\delta p = \frac{1}{3G_0} \left\{ 2G_0 \tilde{\mathbf{N}} : \delta \boldsymbol{\varepsilon} - \left[\frac{dR}{dr} + \frac{\frac{3}{2}k_1 - k_2(\tilde{\mathbf{N}} : \mathbf{X}_n)}{(1 + k_2\Delta r)^2} \right] \delta r \right\} \quad (67)$$

The variation of $\tilde{\mathbf{s}}$ is found by analogy with (51). Using (62), the variation of the *effective stress* $\tilde{\boldsymbol{\sigma}}$ is found to be $\delta \tilde{\boldsymbol{\sigma}} = \tilde{\mathbf{H}}^{\text{alg}} : \delta \boldsymbol{\varepsilon}$, where

$$\begin{aligned} \tilde{\mathbf{H}}^{\text{alg}} = & \mathbf{E}_0 - (2G_0)^2 \frac{\Delta p}{(1 + \frac{3}{2}g)} \frac{\partial^2 f}{\partial \tilde{\boldsymbol{\sigma}} \partial \tilde{\boldsymbol{\sigma}}} - \frac{2}{3} \tilde{\mathbf{N}} \otimes (2G_0 \tilde{\mathbf{N}}) \\ & + \left\{ \frac{2}{3} \left[\frac{dR}{dr} + \frac{\frac{3}{2}k_1 - k_2 \tilde{\mathbf{N}} : \mathbf{X}_n}{(1 + k_2\Delta r)^2} \right] \tilde{\mathbf{N}} - \frac{3G_0}{J_2(\tilde{\boldsymbol{\sigma}} - \mathbf{X})} \frac{\Delta p}{(1 + \frac{3}{2}g)} \frac{k_2}{(1 + k_2\Delta r)^2} \right. \\ & \left. \times \left[\mathbf{X}_n - \frac{2}{3}(\tilde{\mathbf{N}} : \mathbf{X}_n) \tilde{\mathbf{N}} \right] \right\} \otimes \frac{\mathbf{n}^{\text{alg}}}{h^{\text{alg}}} \end{aligned} \quad (68)$$

The variation of the true stress $\boldsymbol{\sigma} = (1 - D)\tilde{\boldsymbol{\sigma}}$ is now found from the equation

$$\delta \boldsymbol{\sigma} = (1 - D)\delta \tilde{\boldsymbol{\sigma}} - \tilde{\boldsymbol{\sigma}} \delta D \quad (69)$$

The variation of the damage D is found by analogy with (52). After some lengthy algebra, it is found that $\delta \boldsymbol{\sigma} = \mathbf{H}^{\text{alg}} : \delta \boldsymbol{\varepsilon}$ where the consistent (or algorithmic) tangent operator can be written in the following compact form:

$$\mathbf{H}^{\text{alg}} = (1 - D)\tilde{\mathbf{H}}^{\text{alg}} - \tilde{\boldsymbol{\sigma}} \otimes \left[\Delta p \tilde{\mathbf{H}}^{\text{alg}} : \frac{\partial y}{\partial \tilde{\boldsymbol{\sigma}}} + \frac{2}{3} y \tilde{\mathbf{N}} - \frac{y}{3G_0} \left[\frac{dR}{dr} + \frac{\frac{3}{2}k_1 - k_2 \tilde{\mathbf{N}} : \mathbf{X}_n}{(1 + k_2\Delta r)^2} \right] \frac{\mathbf{n}^{\text{alg}}}{h^{\text{alg}}} \right] \quad (70)$$

When the plasticity increment is very small, i.e. when $\Delta p \rightarrow 0$, then it can be checked that $\mathbf{H}^{\text{alg}} \rightarrow \mathbf{H}$, where $\mathbf{H}$ is the ‘continuum’ tangent modulus given by (30). Of course, the use of the continuum instead of the consistent tangent operator has no effect on the accuracy of the solution, but it may dramatically alter the convergence of the global Newton-type equilibrium iteration scheme,⁴ especially for large increments (large Δp) with rapidly changing conditions (non-negligible second-order terms).

4.1. Study of the non-damaged case

If damage is inactive during the time increment, then the following simplifications hold: $\Delta D = 0$, $D = D_n$, $y = 0$, $\partial y / \partial \tilde{\sigma} = 0$. If, in addition, D_n is zero, then $p = r$, $\tilde{\sigma} = \sigma$. In that case, the consistent tangent operator simplifies to

$$\mathbf{H}^{\text{alg}} = \mathbf{H}^{\text{mod}} - (2G_0)^2 \frac{\Delta p}{(1 + \frac{3}{2}g)} \frac{\partial^2 f}{\partial \sigma \partial \sigma} - \frac{3G_0}{J_2(\tilde{\sigma} - \mathbf{X})} \frac{\Delta p}{(1 + \frac{3}{2}g)} \frac{k_2}{(1 + k_2 \Delta p)^2} \left[\mathbf{X}_n - \frac{2}{3}(\tilde{\mathbf{N}} : \mathbf{X}_n) \tilde{\mathbf{N}} \right] \otimes \frac{(2G_0 \mathbf{N})}{h^{\text{alg}}} \quad (71)$$

where h^{alg} is the denominator in (59) and $\mathbf{H}^{\text{mod}}$ is the 'modified' continuum tangent modulus defined by

$$\mathbf{H}^{\text{mod}} = \mathbf{E}_0 - \frac{(2G_0 \mathbf{N}) \otimes (2G_0 \mathbf{N})}{h^{\text{alg}}} \quad (72)$$

If $k_2 = 0$ (kinematic hardening is linear, or absent), then further simplifications occur:

$$\mathbf{H}^{\text{alg}} = \mathbf{H} - (2G_0)^2 \frac{\Delta p}{1 + \frac{3}{2}g} \frac{\partial^2 f}{\partial \sigma \partial \sigma} \quad (73)$$

where $\mathbf{H}$ is the *continuum* tangent operator given by equation (30), after setting the damage terms D and $\partial F_D / \partial Y$ to zero. This *simple and explicit* relation shows clearly the *extra term* obtained when an exact linearization of the integration algorithm is performed. Two quantities are particularly relevant: the plasticity increment Δp , and $\partial^2 f / \partial \sigma \partial \sigma$.

5. ANALYTICAL SOLUTIONS FOR LOADINGS WITH CONSTANT TRIAXIALITY RATIO

In this section, we develop analytical solutions for loadings for which the triaxiality ratio, and therefore the triaxiality term, R_v , are constant. Examples are: uniaxial tension ($R_v = 1$) and simple shear ($R_v = \frac{2}{3}(1 + \nu)$). Linear isotropic hardening ($R = kr$) and ductile damage are considered; kinematic hardening is absent.

Using equation (22a) and the yield condition, the damage evolution equation (20) can be rewritten as

$$(1 - D) \dot{D} = \frac{\sigma_y^2}{2E_0 S_0} \left(1 + \frac{k}{\sigma_y} r \right)^2 R_v \dot{r} \quad (74)$$

Time integration of this equation with the initial conditions $r_0 = D_0 = 0$ gives

$$(1 - D)^2 = 1 - \frac{\sigma_y^2}{3E_0 S_0} \frac{\sigma_y}{k} \left[\left(1 + \frac{k}{\sigma_y} r \right)^3 - 1 \right] R_v \quad (75)$$

Large dots in Figures 1(a)–(f) correspond to the transition from strain hardening to strain softening. The material data is that of common steel.

Figure 1(a) shows the evolution of the damage, D , as a function of the isotropic hardening strain, r . From equation (75) it is deduced that for a given r , the value of D is higher for a higher

R_v . This is illustrated in Figure 1(a) which shows that for a given r , a tension test is *more damaging* than a shear test. Taking the derivative of D with respect to r , one finds

$$\frac{dD}{dr} = \frac{1}{(1-D)} \frac{\sigma_y^2}{2E_0 S_0} \left(1 + \frac{k}{\sigma_y} r\right)^2 R_v > 0 \quad (76)$$

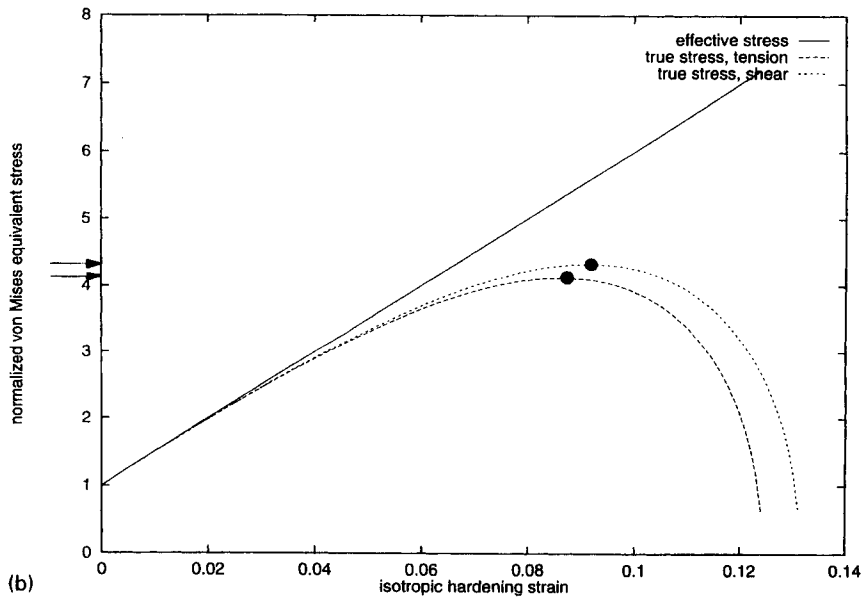
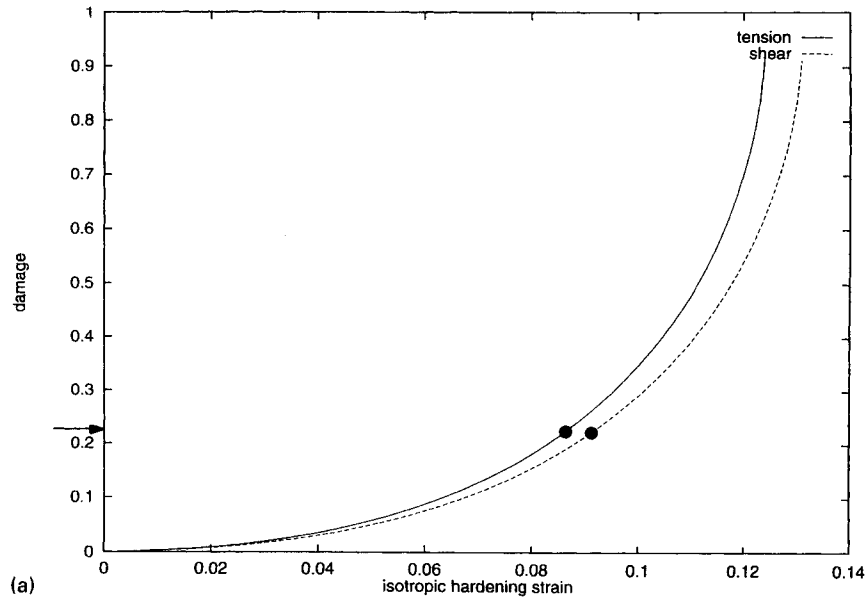


Figure 1(a, b)

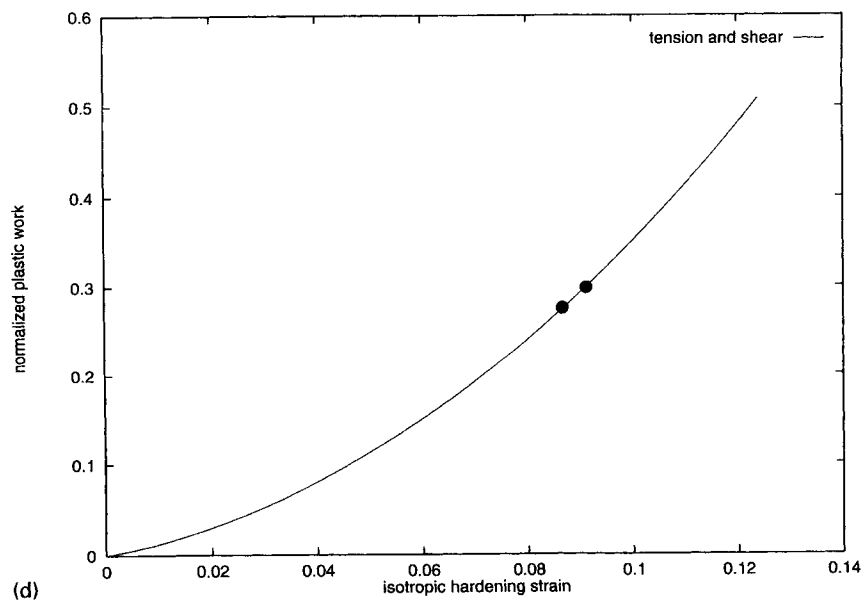
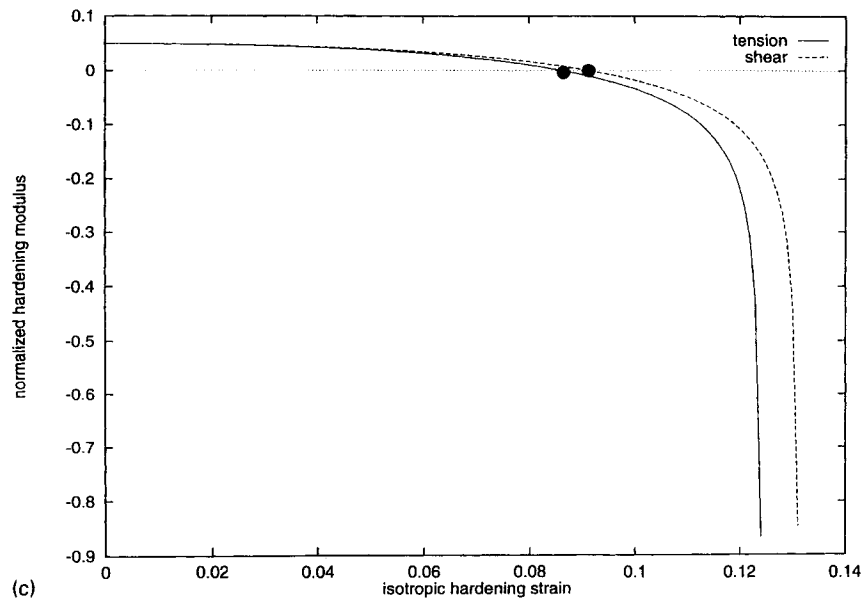


Figure 1 (c, d)

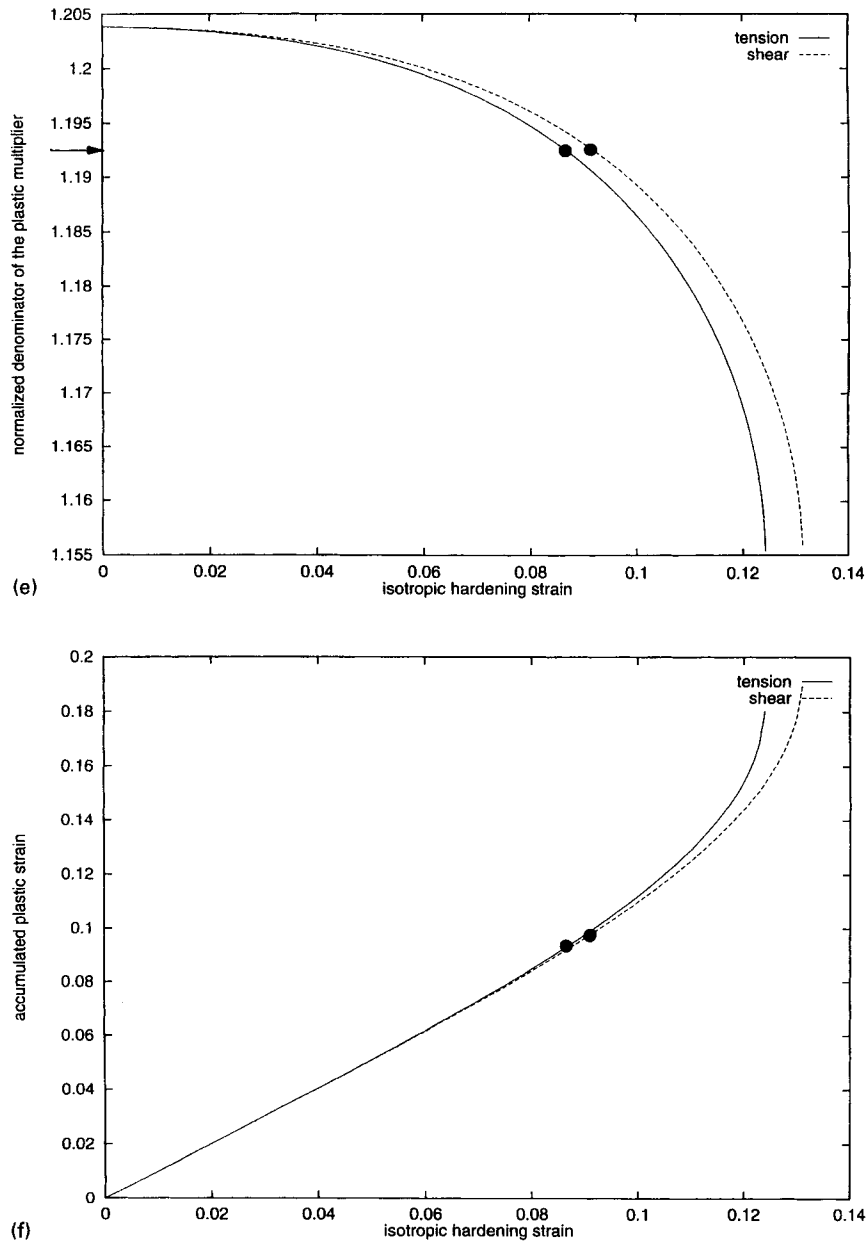


Figure 1. Analytical solutions for loadings with constant triaxiality ratio. Material parameters: $E_0 = 200$ GPa, $\nu = 0.3$, $\sigma_y = 200$ MPa, $k = 10$ GPa, $n = 1$, $S_0 = 0.5$ MPa. Variation of various variables as a function of the isotropic hardening strain, r . (a) Damage, D ; (b) true and effective von mises equivalent stresses, divided by σ_y ; (c) hardening modulus, h^* , divided by E_0 ; (d) plastic work, w^p , divided by σ_y ; (e) denominator of the plastic multiplier, h , divided by E_0 ; (f) accumulated plastic strain, p

The slope of D at the origin is given by

$$\frac{dD}{dr}(r=0) = \frac{\sigma_y^2}{2E_0S_0} R_v \quad (77)$$

The slope of D is *infinite* for $D = 1$, and corresponds to the following value of r :

$$r^{\text{fail}} = \frac{\sigma_y}{k} \left[\left(1 + \frac{3}{R_v} \frac{E_0 S_0}{\sigma_y^2} \frac{k}{\sigma_y} \right)^{1/3} - 1 \right] \quad (78)$$

A numerical application gives $r^{\text{fail}} = 12.43$ per cent in tension and $r^{\text{fail}} = 13.14$ per cent in shear. These failure values correspond to vertical slopes in Figure 1(a).

Using the yield condition, the von Mises equivalent stress is found to be

$$J_2(\sigma) = (1 - D)(\sigma_y + R) = (1 - D)\sigma_y \left(1 + \frac{k}{\sigma_y} r \right) \quad (79)$$

Using equation (75) an analytical expression of $J_2(\sigma)$ as a function of r is found. Figure 1(b) shows $J_2(\sigma)$ vs. r . The *softening* behaviour is illustrated in that figure. Also in Figure 1(b), the 'effective' von Mises equivalent stress $J_2(\tilde{\sigma})$ is plotted versus r . The curve is a straight line with a *positive* slope, which is explained by the fact that as damage increases, the resisting areas decrease, and therefore the effective stresses increase. The straight line in Figure 1(b) would also represent the true von Mises equivalent stress $J_2(\sigma)$, if damage were inactive.

As proved in Section 2.3, the plastic tangent operator, $\mathbf{H}$, becomes *singular* when the hardening modulus h^* is zero. This was also found to be a condition for *softening* for the model under consideration. Using the expressions of F_D , $\tilde{\mathbf{N}}$ and J_2 , relation (37) becomes

$$h^* = (1 - D) \frac{dR}{dr} - \frac{1}{(1 - D)} \frac{Y}{S_0} J_2(\tilde{\sigma}) \quad (80)$$

or, using the damage evolution equation (20) together with $f = 0$,

$$h^* = (1 - D) \frac{dR}{dr} - (\sigma_y + R) \frac{dD}{dr} \quad (81)$$

Taking the derivative of $J_2(\sigma)$ with respect to r in equation (79a) one finds

$$h^* = \frac{dJ_2(\sigma)}{dr} \quad (82)$$

This equality shows that $h^*(r)$ is the *slope* of the curve $J_2(\sigma)$ versus r and that the condition $h^* = 0$ corresponds to the *peak* of that curve. Figure 1(c) shows the variation of h^* as a function of r .

Using equations (81) and (76), it is found that the condition $h^* = 0$ is equivalent to the following value of damage:

$$D^{\text{soft}} = 1 - \sqrt{\frac{3}{5} + \frac{\sigma_y^2}{5E_0S_0} \frac{\sigma_y}{k} R_v} \quad (83)$$

A numerical application gives $D^{\text{soft}} = 0.22$ for *both* tension and shear. Note that this critical damage value is *much smaller* than the ultimate failure value of 1. It is shown in Reference 6 that while r^{soft} is strongly dependent on the triaxiality term, D^{soft} is practically *independent* on R_v . Using (83) and (75) it is found that softening occurs for the following value of the isotropic

hardening strain:

$$r^{\text{soft}} = \frac{\sigma_y}{k} \left\{ \left[\frac{2}{5} + \frac{6}{5R_v} \frac{E_0 S_0}{\sigma_y^2} \frac{k}{\sigma_y} \right]^{1/3} - 1 \right\} \quad (84)$$

A numerical application gives $r^{\text{soft}} = 8.63$ per cent in tension and $r^{\text{soft}} = 9.15$ per cent in shear. Combining equations (84) and (78) gives the following simple relations between softening and failure values of r :

$$r^{\text{soft}} = \left(\frac{2}{5}\right)^{1/3} r^{\text{fail}} - [1 - \left(\frac{2}{5}\right)^{1/3}] \frac{\sigma_y}{k} \approx 0.74 r^{\text{fail}} - 0.26 \frac{\sigma_y}{k} \quad (85)$$

We now compute the value of the von Mises equivalent stress at softening. Using the yield condition, equation (79b), we obtain

$$\frac{J_2(\boldsymbol{\sigma}^{\text{soft}})}{\sigma_y} = (1 - D^{\text{soft}}) \left(1 + \frac{k}{\sigma_y} r^{\text{soft}} \right) \quad (86)$$

A numerical application gives $J_2(\boldsymbol{\sigma}^{\text{soft}})/\sigma_y = 4.12$ in tension and 4.35 in shear.

The increment of *plastic work* is defined by $dw^p = \boldsymbol{\sigma} : d\boldsymbol{\epsilon}^p$. After some algebraic manipulation, this is found to be

$$dw^p = J_2(\boldsymbol{\sigma}) dr = \sigma_y \left(1 + \frac{k}{\sigma_y} r \right) dr \quad (87)$$

Time integration of equation (87) with zero initial conditions gives the following expression for the plastic work, which is independent of the triaxiality ratio:

$$\frac{w^p}{\sigma_y} = \frac{1}{2} \frac{\sigma_y}{k} \left[\left(1 + \frac{k}{\sigma_y} r \right)^2 - 1 \right] \quad (88)$$

Figure 1(d) gives w^p as a function of r . Using equation (85), the following simple relations are found, which are *independent* of any parameter or variable:

$$\frac{(dw^p/dr)^{\text{soft}}}{(dw^p/dr)^{\text{fail}}} = \left(\frac{2}{5}\right)^{1/3} \approx 0.74 \quad (89)$$

The denominator of the plastic multiplier, h , is given by the following expression:

$$h = 3G_0 + (1 - D)k > 0 \quad (90)$$

As mentioned in Section 2.1, when non-linear kinematic hardening is absent, h is guaranteed to be positive. The curve h vs. r is plotted in Figure 1(e). It is seen that after softening occurs, h decreases rapidly, with a vertical slope at failure.

Figure 1(f) shows the variation of the accumulated plastic strain, p , as a function of the isotropic hardening strain, r . Before softening, p and r are almost equal, but in the softening range, p increases very rapidly, with a vertical slope near failure. Let us recall that $p = r$ when damage is inactive.

6. NUMERICAL SIMULATIONS

The constitutive equations presented in Section 2 have been implemented in the general purpose finite element program SPECTRUM²⁷ using the algorithm developed in Sections 3 and 4. Note that several *special* cases, corresponding to different models, can be treated with the same source

code, by setting appropriate values to different material parameters (in SPECTRUM, however, each special case is coded separately, for performance reasons).

- (a) Damage can be turned off by setting the damage threshold, p_D , to a very large positive value.
- (b) Kinematic hardening can be turned off by setting $k_1 = 0$ and $k_2 = 0$.
- (c) Kinematic hardening can be made linear by setting $k_2 = 0$.
- (d) Isotropic hardening can be turned off by choosing power law and $k = 0$ or exponential law and $R_\infty = 0$.
- (e) Isotropic hardening can be made linear by choosing power law and $n = 1$.
- (f) Plasticity can be turned off all together by setting the initial yield stress, σ_y , to a very large positive value.

Note also that two cases can be treated without any modification to the algorithm presented in Sections 3 and 4.

- (i) Isotropic hardening models of the form $R = R(r)$, where R is an arbitrary differentiable scalar function of r .
- (ii) Ductile damage models of the form $\dot{D} = y(\tilde{\sigma}) \dot{p}$, where y is an arbitrary differentiable scalar function of $\tilde{\sigma}$.

6.1. Ductile failure under uniaxial tension

A uniaxial tension test (σ_{11} the only non-zero stress component) is simulated by submitting a cube to normal displacements on one of its faces. The material data are that of common steel. The prescribed displacement increases linearly with time. Equal-size load increments are considered, each corresponding to a total strain increment $\Delta\epsilon_{11} = 1$ per cent, which is *ten times* the initial yield strain ($\sigma_y/E_0 = 0.1$ per cent). The robustness of the proposed algorithm is illustrated in Figures 2(a) and 2(b) where the numerical results are plotted together with the analytical solutions developed in Section 5. The numerical accuracy can be improved by choosing smaller load increments. This is illustrated in Figures 2(a) and 2(b) where the results of a simulation corresponding to $\Delta\epsilon_{11} = 0.5$ per cent are plotted.

6.2. Ductile failure under simple shear

A simple shear test ($\sigma_{12} = \sigma_{21}$ the only non-zero stress components) is simulated by submitting a cube to tangential displacements on one of its faces. The material data are that of common steel. The prescribed displacement increases linearly with time. Equal-size load increments are considered, each corresponding to a total strain increment $\Delta\epsilon_{12} = 1$ per cent (i.e. an engineering shear strain increment $\Delta\gamma_{12} = 2$ per cent), which is *ten times* the initial yield strain ($\sigma_y/E_0 = 0.1$ per cent). The robustness of the proposed algorithm is illustrated in Figures 3(a) and 3(b) where the numerical results are plotted together with the analytical solutions developed in Section 5. The numerical accuracy can be improved by choosing smaller load increments. This is illustrated in Figures 3(a) and 3(b) where the results of a simulation corresponding to $\Delta\epsilon_{12} = 0.5$ per cent are plotted.

6.3. Cyclic elasto-plasticity with homogeneous initial conditions

A cyclic tension-compression strain history in the z -direction is applied to a SPECTRUM 8-node brick element. Non-linear isotropic and kinematic hardenings are considered without

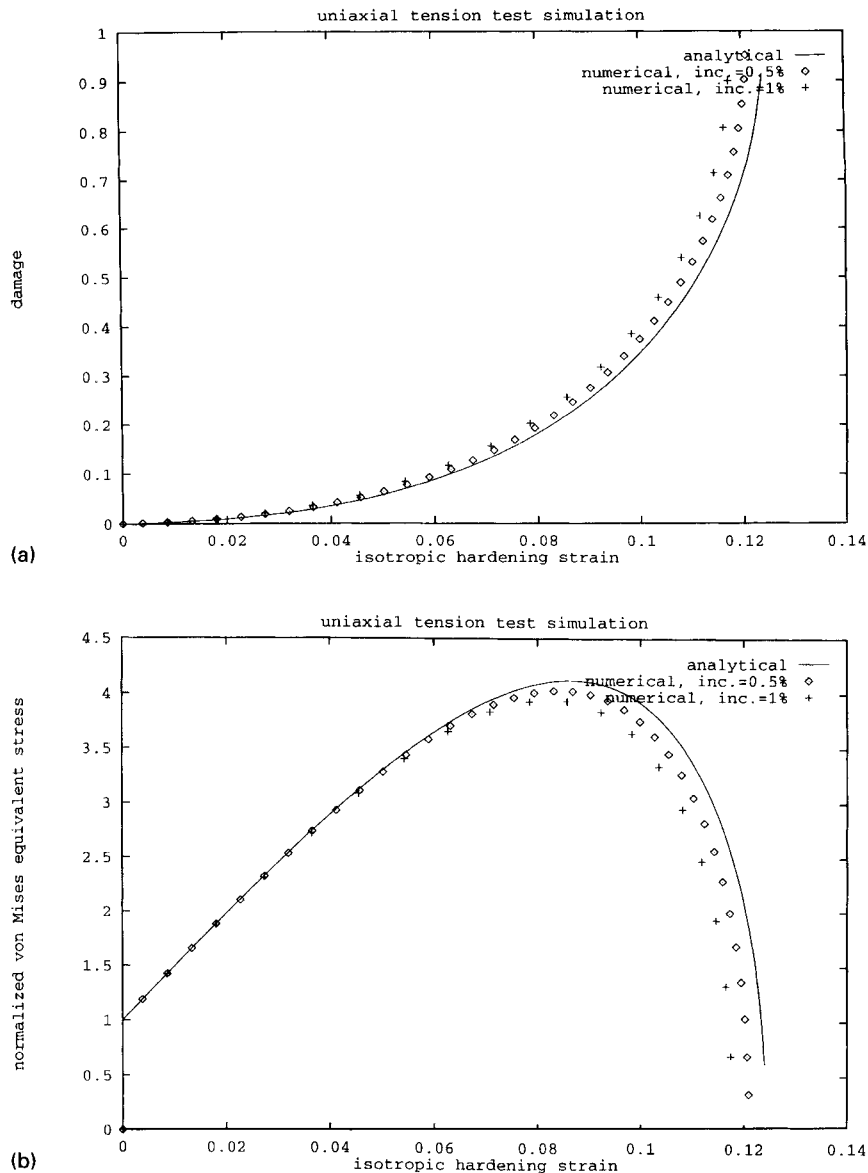


Figure 2. Ductile failure under uniaxial tension. Material parameters are as in Fig. 1. Variation of two variables as a function of the isotropic hardening strain, r : (a) damage, D ; (b) von Mises equivalent stress, $J_2(\boldsymbol{\sigma})$, divided by σ_y

damage evolution and with homogeneous initial conditions. The material is a low-carbon (AISI 1010) steel in a laminated state. The prescribed total strain history is shown in Figure 4(a). Ten cycles were applied, and each one of them was subdivided into 4 equal-size 'time' increments. Figure 4(b) shows the resulting stress-strain loops. The effect of the recall term of the non-linear kinematic hardening evolution law is illustrated by the stabilization of the stress-strain loops (Figure 4(b)) due to the vanishing mean value of the kinematic hardening (Figure 4(c)). Note also that although the xx and yy components of the 'back' stress have different initial values, they

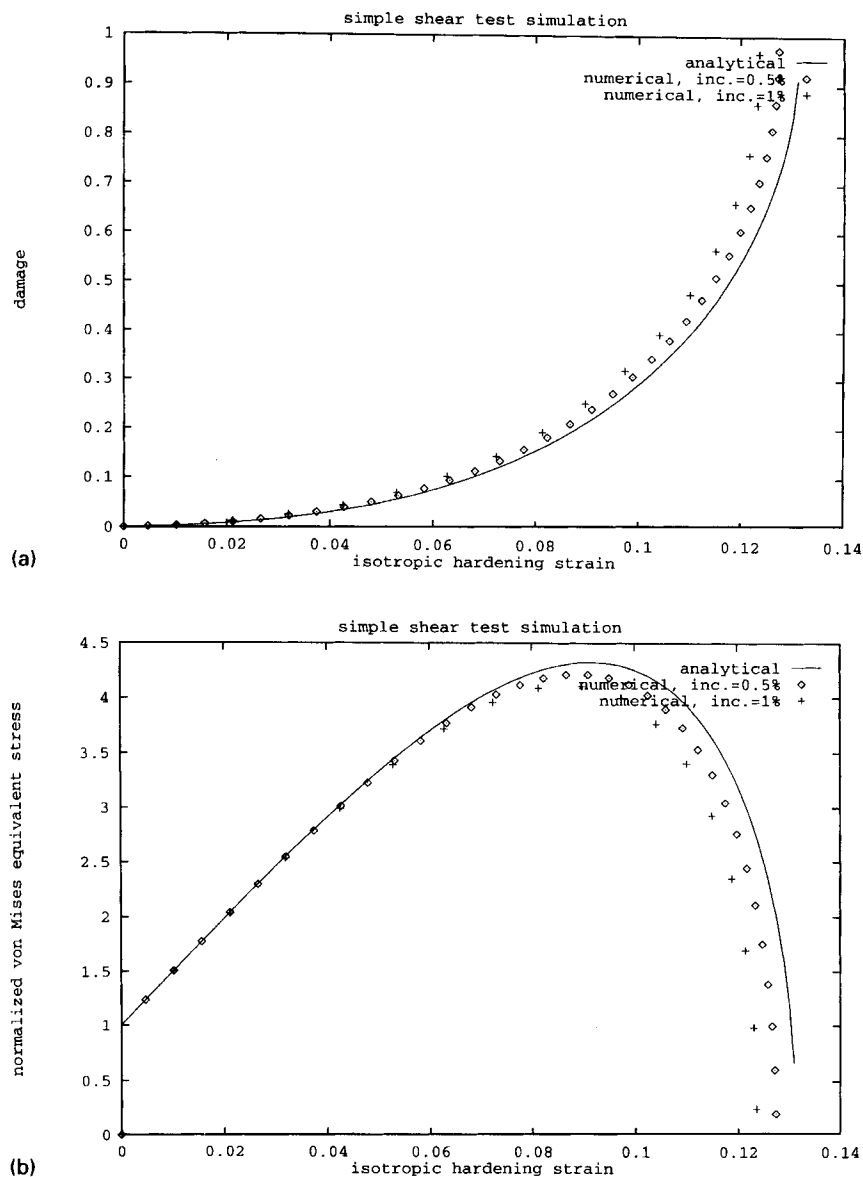


Figure 3. Ductile failure under simple shear. Material parameters are as in Fig. 1. Variation of two variables as a function of the isotropic hardening strain, r : (a) damage, D ; (b) von Mises equivalent stress, $J_2(\sigma)$, divided by σ_y

become equal after the first cycle (which restores the symmetry of the problem, since this is a uniaxial stress state).

6.4. Experimental and numerical study of a notched plate with a macro-defect

The geometry of a flat 2 mm-thick notched specimen is given in Figure 5. A through hole 1 mm in diameter in the vicinity of the notch plays the role of a macro-defect or stress concentrator. The

specimen is made of 2024 aluminium alloy and is subjected to monotonically increasing displacements through bearings which allow free in-plane rotation of each head of the specimen. The original square grid marked on the specimen consists in points 0.18 mm in diameter at a mean distance of 0.50 mm.

For the numerical simulation of the experiment, a plane stress assumption is made. The mesh used and the boundary conditions are shown in Figure 5. SPECTRUM 4-node shell elements are used. The material behaviour is modelled by the constitutive equations of Section 2 with the

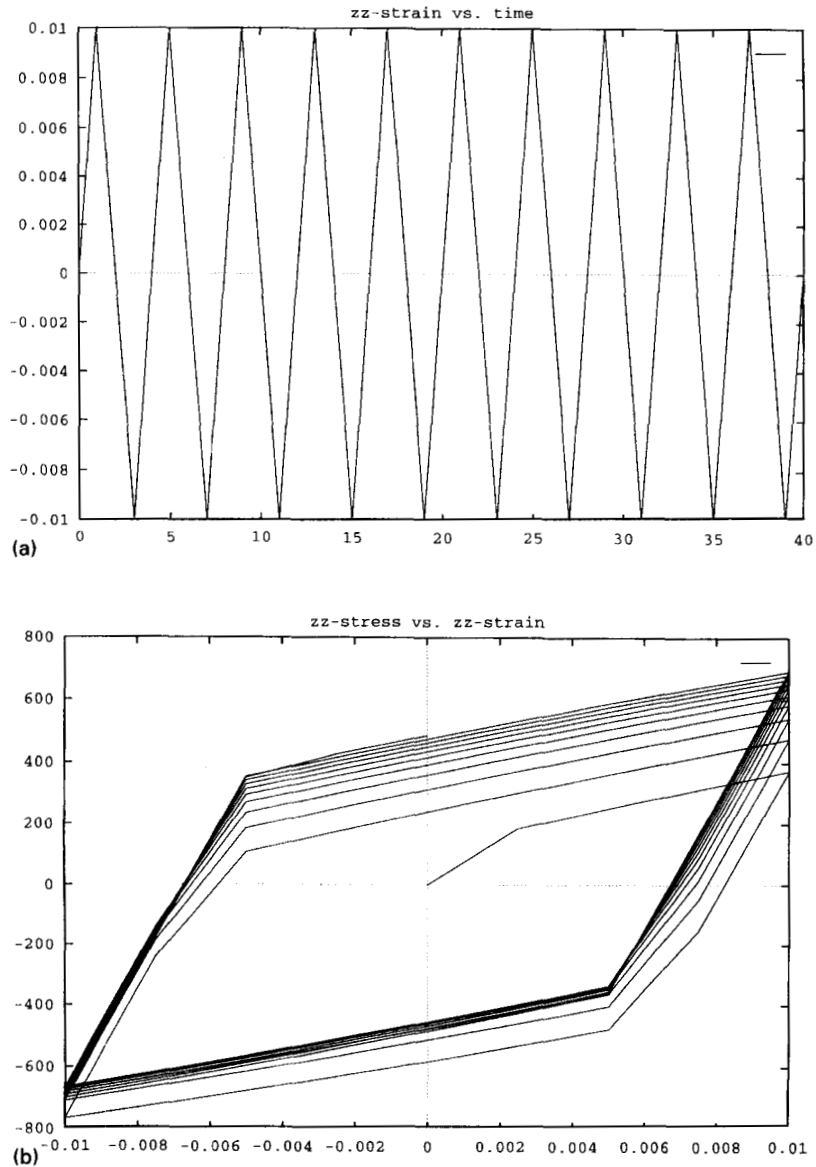


Figure 4(a, b)

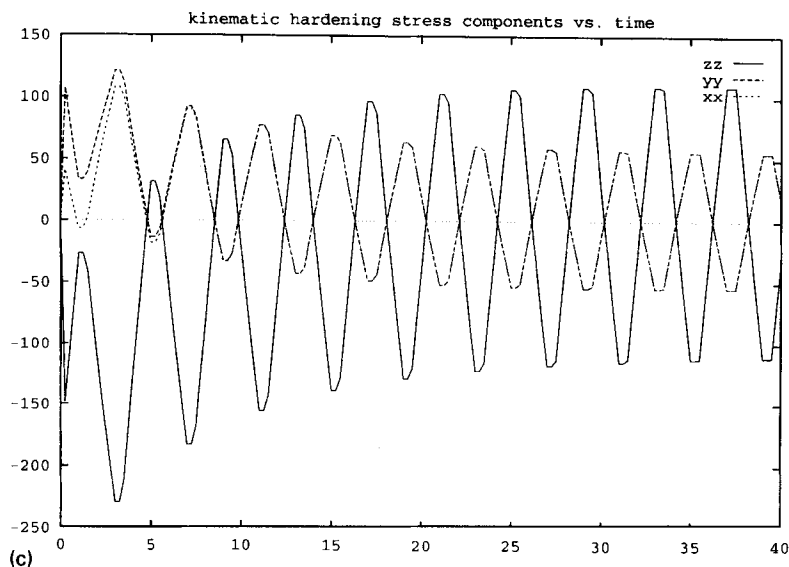


Figure 4. Cyclic elasto-plasticity with homogeneous initial conditions. Material parameters: $E_0 = 210$ GPa, $\nu = 0.3$, $\sigma_y = 200$ MPa, $R_\infty = 2$ GPa, $n = 0.26$, $k_1 = 17$ GPa, $k_2 = 21$. Initial conditions: $r_0 = 0.43$, $(X_{xx})_0 = 53$ MPa, $(X_{yy})_0 = 128$ MPa, $(X_{zz})_0 = -181$ MPa. (a) zz -strain vs. time; (b) zz -stress vs. zz -strain; (c) kinematic hardening stress components vs. time

following features: exponential law isotropic hardening, non-linear kinematic hardening and ductile damage. The behaviour under uniaxial tension, and the material parameters are given in Figure 6.

Figure 7 shows a good agreement between the experimentally measured and the numerically computed load–displacement curves. We note the good accuracy with which the maximum load is predicted. That load corresponds to the first macro-crack, which initiates at the hole and towards the notch. More numerical and experimental results concerning strain localization and cracking are given in Reference 6.

Figures 8(a)–(f) give some numerical responses at the most damaged point. Figure 8(a) shows that the peak in the global load–displacement curve corresponds, *locally*, to the *peak* in the stress–strain curve, in the most damaged point. Figure 8(b) shows that the critical value of damage is $D^{\text{crit}} \approx 0.2 \approx D^*$, where D^* corresponds to the peak of the stress–strain curve in a *uniaxial* tension test. This confirms a result in Section 5 and in Reference 6 which states that the value of damage for which softening occurs is independent on the stress state, and can therefore be identified from a uniaxial tension test. Figure 8(d) shows that the triaxiality ratio—which is equal to $\frac{1}{3}$ in uniaxial tension—has a mean value of 0.48 and peaks up as the critical stage is approached.

7. CONCLUSIONS

This paper dealt with a class of rate-independent metal plasticity models which exhibit non-linear isotropic hardening, non-linear kinematic hardening (Chaboche–Marquis model) and ductile damage (Lemaitre–Chaboche model). This paper is self-contained; it enables efficient implementation of the constitutive equations using the proposed algorithmic procedure. In Section 2, the constitutive equations were presented and studied. Several results which are useful for numerical

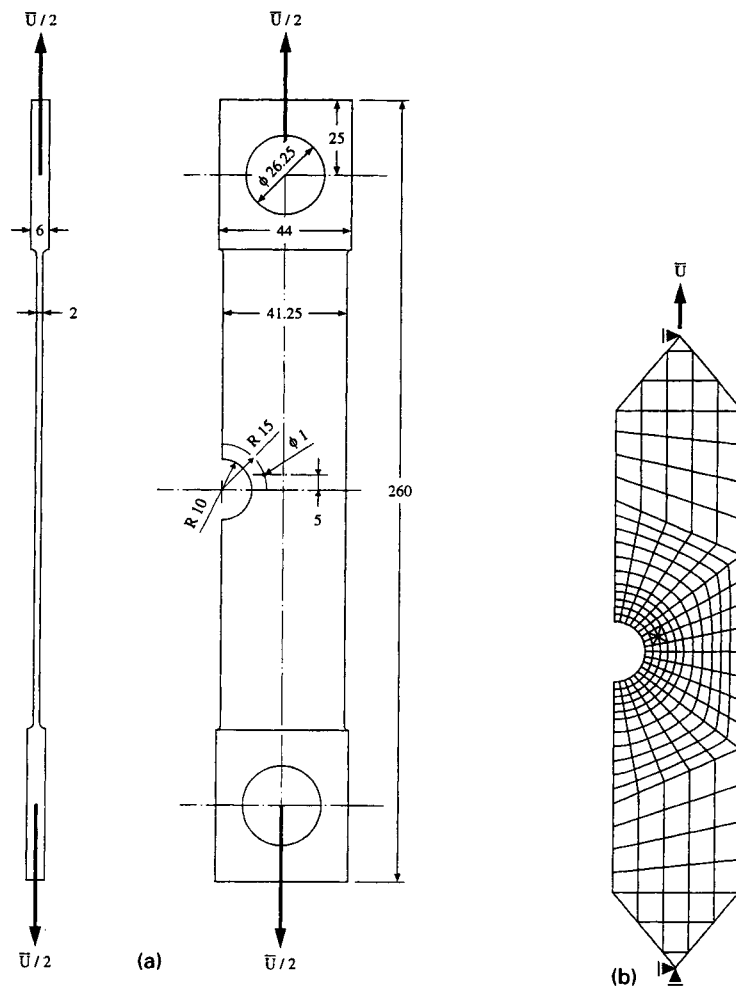


Figure 5. Notched plate with a macro-defect: (a) geometry and loading, (b) mesh and boundary conditions

analysis (plastic multiplier, loading/unloading conditions, hardening/softening criteria, tangent operator, singularity condition, etc.) were derived and discussed. In Section 3, a fully implicit integration algorithm was developed, and in Section 4, the corresponding algorithmic tangent operator was derived by consistent linearization of the discretized constitutive equations. Despite the complexity of the constitutive equations, closed-form expressions were derived in Sections 3 and 4, without any approximations. In the final expressions, there is no need to solve linear systems or to invert matrices, as in other references. In Section 5, analytical results for loadings with constant triaxiality ratio, were developed and discussed. Those results provide additional insight to the constitutive model, and can be used to check the numerical implementation. In Section 6, four numerical applications are presented. The first two simulate ductile failure under uniaxial tension and simple shear, respectively. The numerical results are compared against analytical results. The third example illustrates the role of the non-linear kinematic hardening model in predicting material response under cyclic loading history. The last example is a

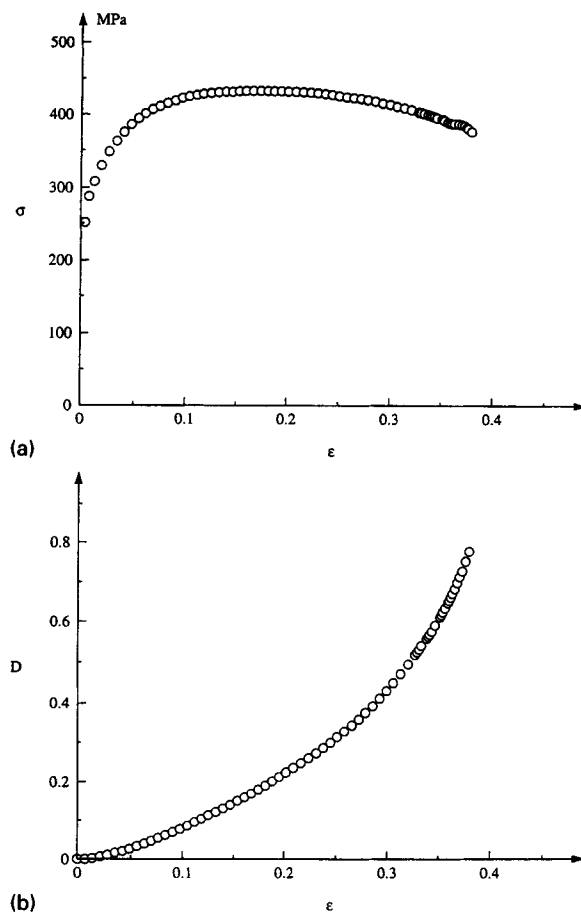


Figure 6. Notched plate with a macro-defect. Material behaviour under uniaxial tension. Material parameters: $E_0 = 72$ GPa, $\nu = 0.33$, $\sigma_y = 273.5$ MPa, $R_\infty = 275$ MPa, $n = 1.86$, $k_1 = 3302$ MPa, $k_2 = 37.2$, $S_0 = 1.3$ MPa. (a) stress vs. strain; (b) damage vs. strain

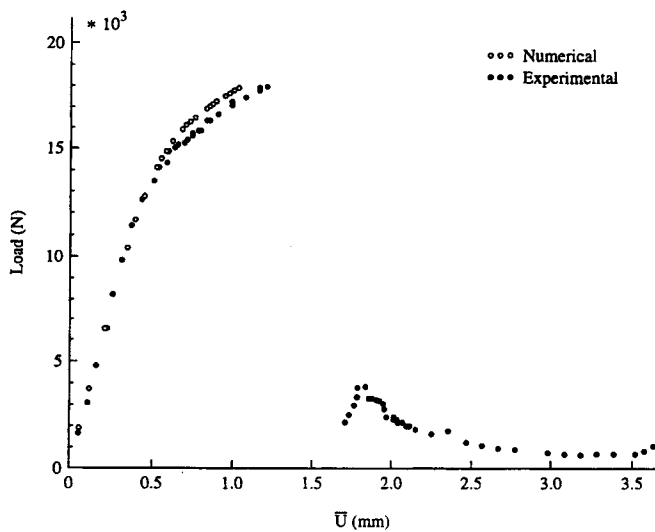


Figure 7. Notched plate with a macro-defect. Experimental and numerical load-displacement curves

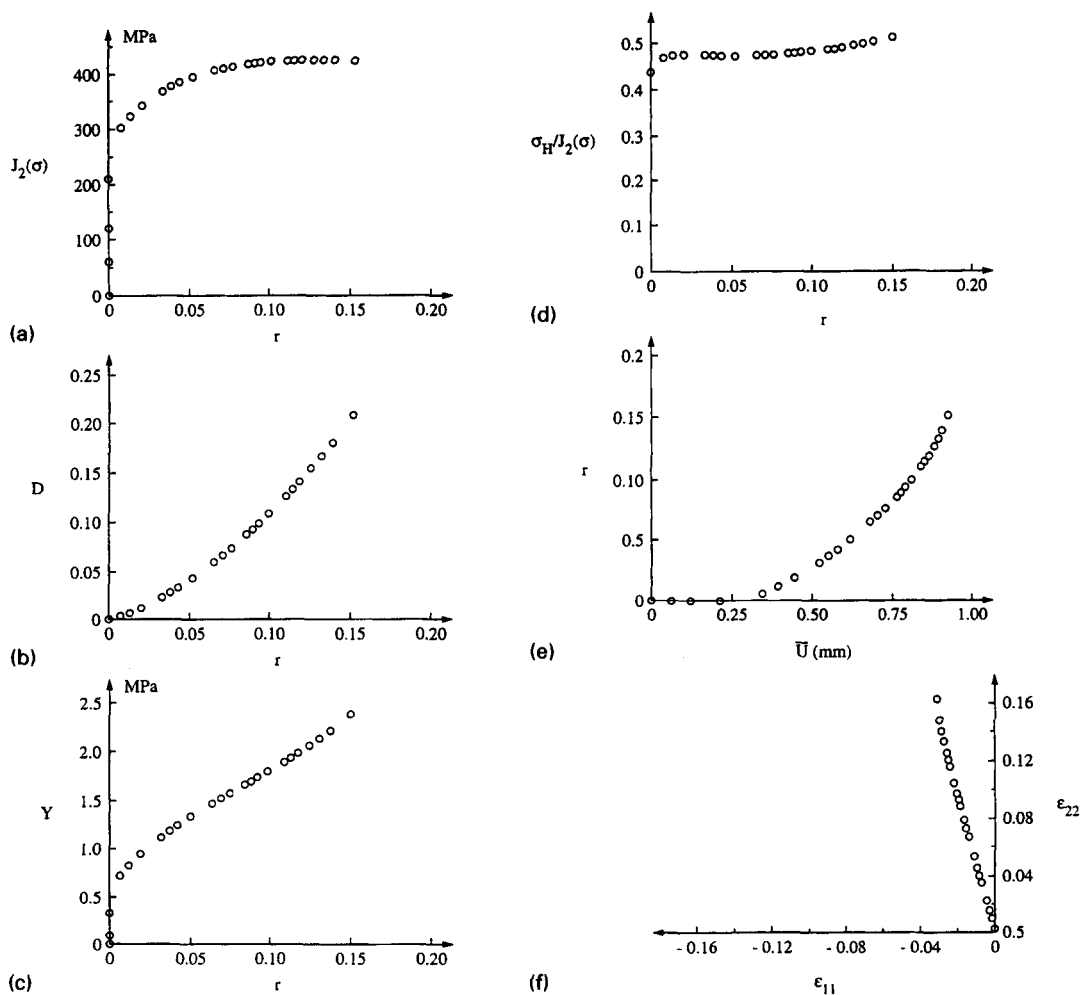


Figure 8. Notched plate with a macro-defect. Local responses at the most damaged point: (a) von Mises equivalent stress vs. r ; (b) damage vs. r ; (c) strain energy release rate vs. r ; (d) triaxiality ratio vs. r ; (e) r vs. global displacement; (f) 22-strain vs. 11-strain

structural problem and concerns a notched plate with a macro-defect. Numerical and experimental results are presented.

Since this paper only dealt with the infinitesimal theory of elasto-plasticity, results which involve rather large strains (say 10 per cent or more) are not reliable. As noted in Reference 28, in the presence of kinematic hardening, classical finite strain implementations—based on hypoelasticity and additive strain rate decomposition—lead to unacceptable behaviour. However, Simo²⁹ has shown that another formulation—based on hyperelasticity and multiplicative deformation gradient decomposition—is free from the spurious oscillations inherent to rate formulations employing the Jaumann derivative. Therefore, it seems that a finite strain implementation of the class of models used in this paper would be based on extending and generalizing Simo's algorithm. This may be the subject of a future work.

The problems of bifurcation and localization induced by the strain softening character of the constitutive equations were not addressed in this paper. For information about those aspects, see Reference 6 and references therein.

Finally, a simplified micro-mechanics version of the damage model is presented in Reference 15 and used to predict ductile failure in one or multidimension, brittle failure, low and high cycle fatigue with the non-linear accumulation, and multiaxial fatigue. A post-processor is developed and its computer listing is provided.

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APPENDIX

Plane stress problems

The algorithm presented in Sections 3 and 4 is 'strain-driven'. However, in a plane stress situation, the out-of-plane strain component is a kinematic unknown.

Let us consider a plane stress problem in the (1, 2) plane. In order to minimize changes to the algorithm developed in Sections 3 and 4, two key ingredients are used. First, 4 components (11, 22, 33, 12) are stored for all symmetric second-order tensors (stress, strain, etc.) Second, the following notation is introduced: $\bar{\delta}_{ij} \equiv I_{ij33}^{\text{dev}} = \delta_{i3}\delta_{j3} - (\frac{1}{3})\delta_{ij}$.

With respect to Section 3, the main difference is that we have an additional unknown: ε_{33} , and an additional equation to be satisfied: $\sigma_{33} = 0$.

Equations (38) are unchanged, with $\varepsilon_{33}^{\text{trial}}$ trivially found from the condition: $\sigma_{33}^{\text{trial}} = 0$.

A term is added to the right-hand side (RHS) of (39): $(\varepsilon_{33} - \varepsilon_{33}^{\text{trial}})(2G_0\bar{\delta} + K_0\mathbf{1})$

A term is added to the RHS of (40a): $3K_0(\varepsilon_{33} - \varepsilon_{33}^{\text{trial}})$. Equation (40b) is unchanged.

A term is added to the RHS of (41a): $2G_0(\varepsilon_{33} - \varepsilon_{33}^{\text{trial}})\bar{\delta}$. The opposite of this term is added to the definition of $\mathbf{k}_s$. Equations (41b, c, d, f) and (42b), (42c) remain unchanged. In (41e) the only change is that: $y = y(\bar{\mathbf{s}}, \varepsilon_{33})$.

There is an additional equation, corresponding to the condition $\sigma_{33} = 0$. It reads:

$$k_{\varepsilon_{33}} \equiv (K_0 + (\frac{4}{3})G_0)\varepsilon_{33} - 2G_0[\varepsilon_{33}^p]_n + \Delta p\tilde{N}_{33} + (K_0 - (\frac{2}{3})G_0)(\varepsilon_{11} + \varepsilon_{22}) = 0$$

When Newton's method is applied, a term is added to the left-hand-side (LHS) of (43a): $-2G_0\bar{\delta}c_{\varepsilon_{33}}$. Equations (43b) and (43c) are unchanged. A term is added to the LHS of (43d): $-\Delta p(\partial y/\partial \varepsilon_{33})c_{\varepsilon_{33}}$.

A new equation is added to the system of equations (43). It reads

$$k_{\varepsilon_{33}} + (K_0 + (\frac{4}{3})G_0)c_{\varepsilon_{33}} - 2G_0\tilde{N}_{33}c_p - 2G_0\Delta p\frac{\partial \tilde{N}_{33}}{\partial \bar{\mathbf{s}}}:(\mathbf{c}_s - \mathbf{c}_x) = 0$$

The procedure detailed in Section 3 is then used to solve the new system of equations (43). The (lengthy) derivations are not given here.

Once the time-integration algorithm has converged, the consistent tangent operator is found. There are no modifications to the algorithm of Section 4, since the variation of the total strain tensor is already accounted for in that section. In the plane stress case, one finds (in matrix notation): $\{\delta\sigma\} = [\mathbf{H}^{\text{alg}}]\{\delta\varepsilon\}$ where $\{\delta\sigma\} = [\delta\sigma_{11} \ \delta\sigma_{22} \ \delta\sigma_{33} \ \delta\sigma_{12}]^T$, and $\{\delta\varepsilon\} = [\delta\varepsilon_{11} \ \delta\varepsilon_{22} \ \delta\varepsilon_{33} \ 2\delta\varepsilon_{12}]^T$.

A 3×3 tangent matrix, which relates $\{\delta\sigma\} = [\delta\sigma_{11} \ \delta\sigma_{22} \ \delta\sigma_{12}]^T$ to $\{\delta\varepsilon\} = [\delta\varepsilon_{11} \ \delta\varepsilon_{22} \ 2\delta\varepsilon_{12}]^T$, is trivially found by imposing the condition $\delta\sigma_{33} = 0$ and eliminating $\delta\varepsilon_{33}$.

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